(From Trivedi)

# SESSION 15

## Regression Model

Statistical relationship between two or more rvs.

Example :

Throughput Y and degree of multiprogramming X

(i) How are these rvs related ? – Correlation Coefficient

(ii) Predict throughput y at a given level of x

→ Dependence of Y on X.

Regression line/curve gives dependence of Y on X.

Conversely : We can also study dependence of X on Y.

PROBLEM :

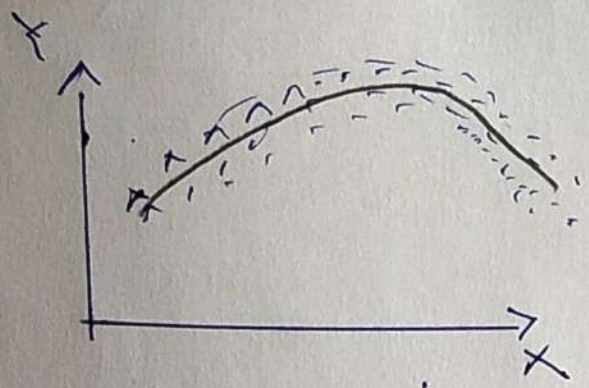
Population Distribution is specified

Find regression curve that "best approximates" the dependence from the sampled data.

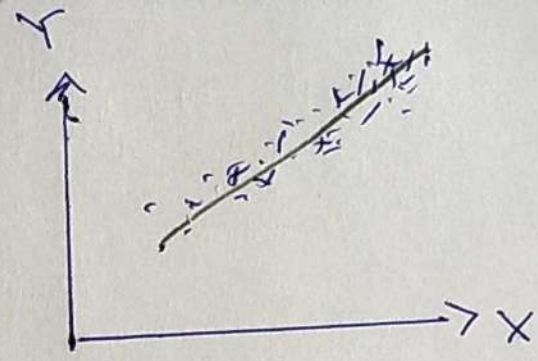
X and Y have joint distn $f(x, y)$

Question Design a fn. $d(x)$ so that rv $d(X)$ is as close as possible to Y

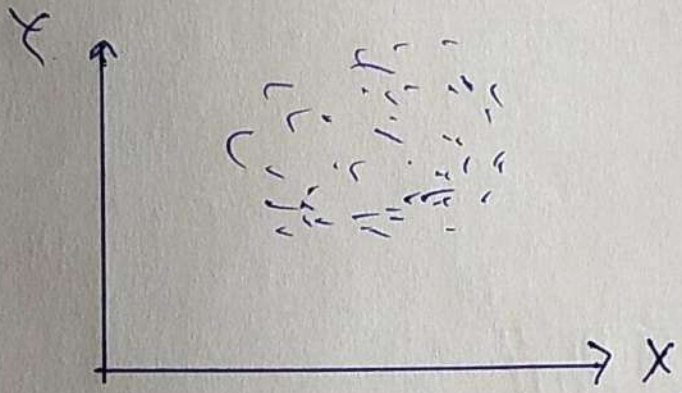
## Scatter Diagram




showing nonlinear dependence




showing linear dependence




No specific dependence

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Modeling error incorporated so that

$$y = d(x) + \epsilon \qquad (1)$$

Reason for introducing error $\epsilon$ — Attempting to simplify the joint distribution of X and Y — Attempt to postulate elementary functional dependence.

Difference is $D = Y - d(X)$ (2)

Choose $d(x)$ for which $E[D^2]$ is minimum is called <u>least-squares regression curve of Y on X</u>

Note $d(x) = E[Y|x]$ (3)

$\uparrow$

* — This requires conditional distributions (In practice more difficult to obtain)

** Accordingly, often we restrict our choices to specific class of functions and minimize $E[D^2]$ over that class

Common Choice: (4)

$$d(x) = a + bx$$

Example

Given $f(x,y) = \left[e^{-x(y+1)}\right]x$, $x \geq 0$, $y \geq 0$

Find regression of Y on X.

Soln (5)

$$f_X(x) = \int_0^\infty f(x,y)\,dy$$

$\uparrow$

Marginal pdf of X

$$\text{or } f_X(x) = \int_0^\infty x\, e^{-x(y+1)}\,dy = x e^{-x}\int_0^\infty e^{-xy}\,dy$$

$$= x e^{-x}\left|\frac{e^{-xy}}{-x}\right|_0^\infty = e^{-x}, \quad x > 0 \quad (6)$$

Conditional pdf of Y on X

$$f(y|x) = \frac{f(x,y)}{f_X(x)} = x e^{-xy} \quad (7)$$

Regression curve of Y given $X = x$:

$$y = E[Y|X=x] = \int_0^\infty y \cdot f(y|x)\,dy$$

$$= \int_0^\infty y\, x e^{-xy}\,dy = \frac{1}{x}.$$

Regression of Y on X is $y = \frac{1}{x}$ (4)

or $xy = 1$ (rectangular hyperbola)

$\boxed{\text{Not linear}}$

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Ex- Find the regression of Y on X when the joint pdf is given by

$$f(x,y) = \frac{y}{(1+x)^4} \exp\left[-\frac{y}{1+x}\right], \quad x, y > 0$$

<u>Soln</u>

$$f_X(x) = \int_0^\infty f(x,y)\,dy = \frac{1}{(1+x)^2}, \quad x \geq 0 \quad (5)$$

and

$$f(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{y}{(1+x)^2} \exp\left[-\frac{y}{1+x}\right], \quad y \geq 0$$

Therefore regression of Y on X is

$$y = E[Y|X=x] = \int_0^\infty y\, f(y|x)\,dy$$

$$= 2(1+x).$$

The regression of Y on X is (6)

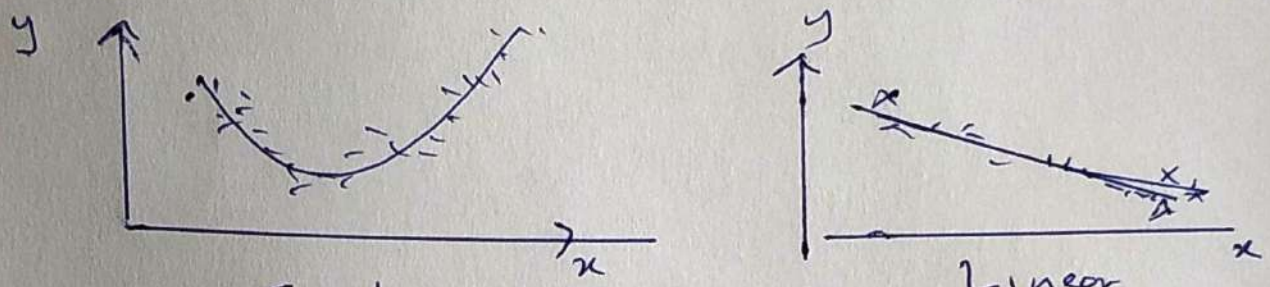
$y = 2(1+x) \leftarrow$ <u>LINEAR</u>

We note
Conditional distribution – Difficult in practice.

– Common choice $d(x) = a + bx$.

– Given $n$ pairs of measurement $(x_1, y_1) \cdots (x_n, y_n)$ in the $(x, y)$ plane (Scatter diagram)

— <u>Approximating Curve</u> ; observations cluttered around some curve in $(x, y)$ plane




Curvilinear regression (Nonlinear)

Linear regression (Linear)

## Linear Model

Let the dependence of $Y$ on $X$ ~~of~~ such that

$$d(x) = a + bx \qquad (7)$$

Choose $a$ and $b$ such that

$$G(a, b) = E[D^2] = E[\{Y - d(x)\}^2] \text{ is minimum} \qquad (8)$$

We have

$$G(a, b) = E[(Y - a - bx)^2]$$

$$= E[(Y - \mu_Y) + (\mu_Y - a) - b(X - \mu_X) - b\mu_X]^2 \qquad (9)$$

Squaring and taking expectation, we get

$$G(a,b) = E[(Y-\mu_Y)^2 + (\mu_Y - a)^2 + b^2(X-\mu_X)^2 + b^2\mu_X^2$$
$$+ 2(Y-\mu_Y)(\mu_Y - a) - 2b(Y-\mu_Y)(X-\mu_X)$$
$$- 2b(Y-\mu_Y)\mu_X - 2b(\mu_Y - a)(X-\mu_X)$$
$$- 2b(\mu_Y - a)\mu_X + 2b^2(X-\mu_X)\mu_X]$$

Since $E[Y-\mu_Y]=0$ and $E[X-\mu_X]=0$, we get

$$G(a,b) = \sigma_Y^2 + b^2\sigma_X^2 + (\mu_Y - a)^2 + b^2\mu_X^2$$
$$- 2b\rho\sigma_X\sigma_Y - 2b\mu_X(\mu_Y - a); \quad \rho = \frac{\mathrm{cov}(X,Y)}{\sigma_X\sigma_Y}$$

or $$G(a,b) = \sigma_Y^2 + b^2\sigma_X^2 + (\mu_X - a - b\mu_X)^2 - 2b\rho\sigma_X\sigma_Y \quad (10)$$

Our objective is to choose a and b so that $G(a,b)$ is minimized.

$$\frac{\partial G}{\partial a} = -2(\mu_Y - a - b\mu_X) = 0 \quad (*)$$

$$\frac{\partial G}{\partial b} = 2b\sigma_X^2 - 2\mu_X\underbrace{(\mu_Y - a - b\mu_X)}_{=0} - 2\rho\sigma_X\sigma_Y = 0 \quad (**)$$

Thus the optimal values of a and b are:

$$\boxed{b = \rho\frac{\sigma_Y}{\sigma_X}} \checkmark \quad \text{and} \quad a = \mu_Y - b\mu_X = \mu_Y - \rho\frac{\sigma_Y}{\sigma_X}\mu_X$$

We find

$$b = \frac{\mathrm{cov}(X,Y)}{\mathrm{Var}\,X}, \quad a = E[Y] - \frac{\mathrm{cov}(X,Y)}{\mathrm{Var}\,X}E[X]. \quad (11)$$

<u>Linear regression curve of Y on X</u>

$$y = E[Y] - \frac{\mathrm{cov}(X,Y)}{\mathrm{Var}\,X}(E(X) - x)$$

Linear regression curve of Y on X

$$\frac{y-\mu_Y}{\sigma_Y} = \rho\left(\frac{x-\mu_X}{\sigma_X}\right) \qquad (12)$$

Similarly linear curve of X on Y can be derived as

$$\frac{x-\mu_X}{\sigma_X} = \rho\left(\frac{y-\mu_Y}{\sigma_Y}\right) \qquad (13)$$

To determine : Square of Minimum Prediction Error

We have obtained (see eq 10)

$$G(a,b) = \sigma_Y^2 + b^2\sigma_X^2 + (\mu_X - a - b\mu_X)^2 - 2b\rho\sigma_X\sigma_Y$$

Also note

$$\frac{\partial G}{\partial a} = 0 \implies (\mu_Y - a - b\mu_X) = 0$$

Therefore $G(a,b) = \sigma_Y^2 + b^2\sigma_X^2 - 2b\rho\sigma_X\sigma_Y$

Substituting $b = \frac{\sigma_Y}{\sigma_X}\cdot\rho$, we have

$$E[D^2] = G(\hat{a},\hat{b}) = (1-\rho^2)\,\sigma_Y^2 \qquad -14-$$

$-1 \le \rho \le 1$,

$E(D^2) = 0$ for $\rho = \pm 1$.

This implies both lines of regression viz Y on X and X on Y coincide. Therefore, X and Y are completely dependent — Have functional relationship with each other.

## Special Case : $\rho = 0$ (Uncorrelated)

Lines of regression $y = \mu_Y$ and $x = \mu_X$ are at right angle.

Recall : X and Y independent then $\rho = 0$
Converse does not hold in general.

* - special case of bivariate normal distribution $\rho = 0$ implies independence

↓

One can use $\rho$ as a measure interdependence in cases of normal or nearly normal.

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Square of minimum prediction error

$$E[D^2] = \sigma_Y^2 - \rho^2 \sigma_Y^2 = \sigma_Y^2 (1 - \rho^2)$$

$$\text{or} \quad \sigma_Y^2 = \underbrace{\rho^2 \sigma_Y^2}_{\text{linear dependence of } Y \text{ on } X} + E[D^2]$$

residual variance not explained by linear relationship

Why this predictor attractive?

↓

Knowledge of first two moments and $\rho$.

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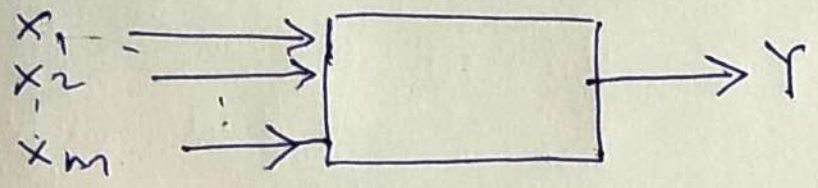
More general Case : Input - Output ~~of~~ behavior of stochastic system with m inputs (independent variables $\underset{\sim}{X} = (X_1, X_2, \cdots X_n)$.




Appropriate predictor of $Y$ is $d(\underset{\sim}{X}, \underset{\sim}{a})$

$d(\underset{\sim}{X}, \underset{\sim}{a})$ is parametrized family of functions with a vector of parameters $\underset{\sim}{a}$.

Generally the form of $d(\underset{\sim}{X}, \underset{\sim}{a})$ is to be inferred – observed data.

<u>Nonlinear Models</u> (More difficult to handle)

$$\frac{\partial}{\partial a_i} E[(Y - d(\underset{\sim}{X}, \underset{\sim}{a}))^2] = 0 \quad \forall i$$

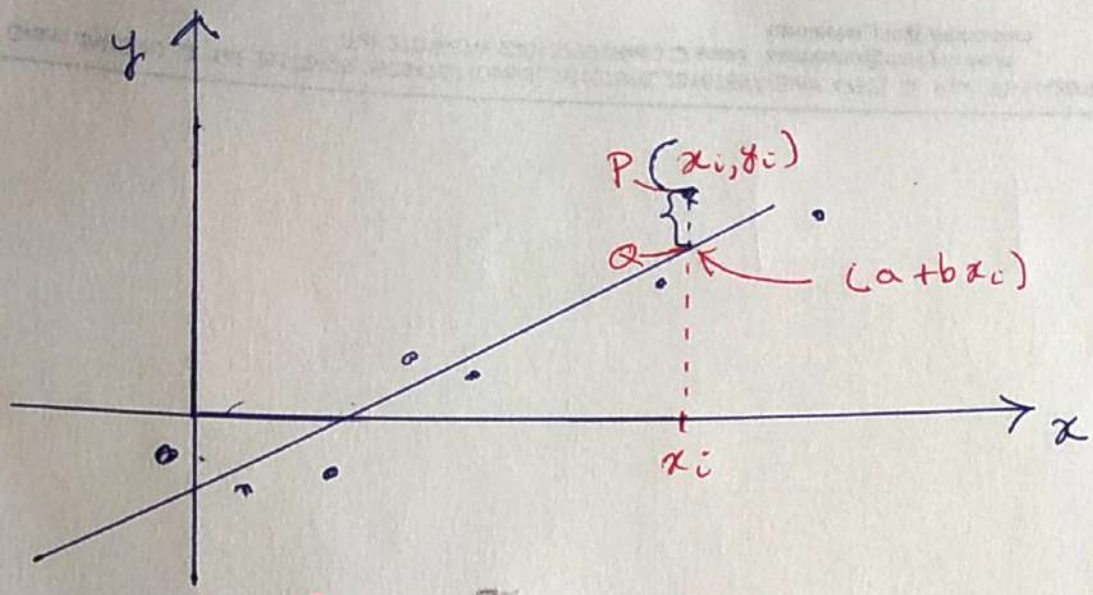
<u>NOTE</u>

Regression function is nonlinear in independent rv $\underset{\sim}{X}$ but could be linear function of the parameters. For example

$$\left.\begin{aligned} d &= a + b \ln x \\ d &= a + b \sin x \end{aligned}\right| \Rightarrow \text{are linear models in } a \text{ and } b.$$

<u>Linear Squares Curve Fitting.</u>

"Best" — line which minimizes the sum of squares of $y$ coordinate deviation from the line




$$PQ = (y_i - a - bx_i)$$

Given points $\{(x_i, y_i) \mid i = 1, 2, \cdots n\}$

Choose a and b so as to minimize the sum of squares of vertical distances between $(x_i, y_i)$ and the line.

$$S = \sum_{i=1}^{n} [y_i - (a + bx_i)]^2 \qquad (17)$$

Minimize S with respect to a and b

We have

$$\frac{\partial S}{\partial a} = -2 \sum_{i=1}^{n} (y_i - a - bx_i) = 0$$

$$\frac{\partial S}{\partial b} = -2 \sum_{i=1}^{n} (y_i - a - bx_i) x_i = 0$$

We obtain the following pair of equations called 'NORMAL' equations

$$n.a + b \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i \qquad (18)$$

$$a \sum_{i=1}^{n} x_i + b \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i y_i$$

By considering ~~ab~~ second derivatives of S, it is easy to show the values of a and b that satisfy normal equations then S is minimized.

## Least Squares line

Let $\hat{a}$ and $\hat{b}$ be defined as

$$\hat{a} = \bar{y} - \hat{b}\,\bar{x}$$

$$\hat{b} = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} \qquad (19)$$

where

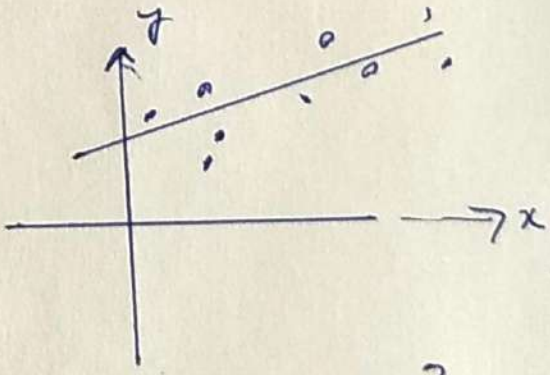
$$\bar{x} = \frac{1}{n}\sum x_i \;,\; \bar{y} = \frac{1}{n}\sum y_i$$

The line defined by $y = \hat{a} + \hat{b}x$ is called the least squares line.

## Probabilistic Counterpart

Consider two discrete rvs X and Y with joint pmf

$$P(X = x_i, Y = y_j) = \frac{1}{n}, \quad i = 1, \dots n$$




Regression line of Y on X

Minimize

$$E[(Y - (a + bX))^2] = \sum_{i=1}^{n} \frac{1}{n}[y_i - (a + bx_i)]^2 \qquad -(20)-$$

Summary

Simple linear regression

$$E[Y \mid X = x] = a + bx \quad \text{for all } x$$

RVs $Y_i$, $i = 1, 2, \dots n$ are of the form

$$\underline{Y_i = a + bx_i + \epsilon_i}, \quad i = 1, 2, \dots n$$

Random variables $\epsilon_1, \epsilon_2, \dots \epsilon_n$ are errors which cause deviations about the linear relationship $a + bx$, $i = 1, \dots n$.

The errors are independent and assumed to be normally distributed

$$E[\epsilon_i] = 0 \quad \text{var}[\epsilon_i] = \sigma_i^2 \quad i = 1, \dots n$$

↑ Recall CLT.

It follows:

$$E[Y_i] = E[Y \mid X = x_i] = E[a + bx_i + \epsilon_i]$$
$$= a + bx_i = \mu_i$$

$$V[Y_i] = V[Y \mid X = x] = V[a + bx_i + \epsilon_i]$$
$$= \text{Var}[\epsilon_i] = \sigma^2$$

$Y_1, \dots Y_n$ are mutually independent because of mutual independence of $\epsilon_1, \epsilon_2 \dots \epsilon_n$.

(From A. O Allen)